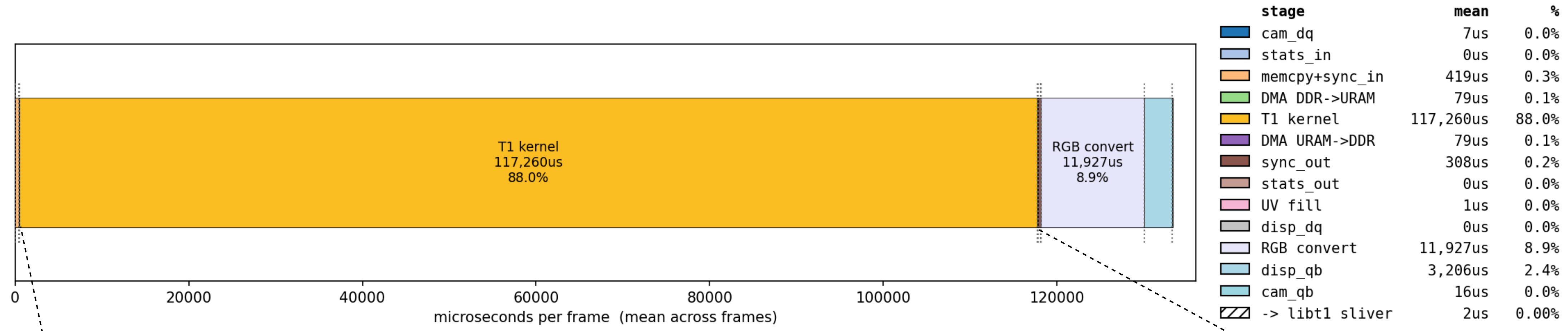
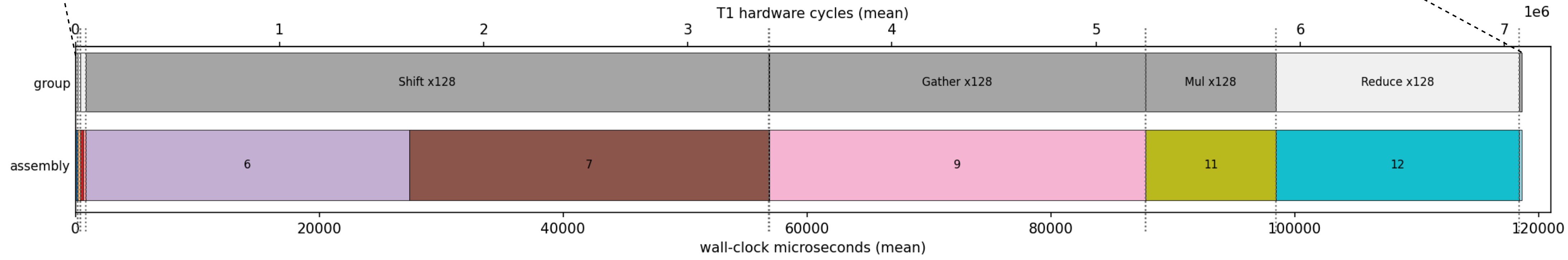


VisionSoC pipeline per-frame breakdown N=99 frames (skipping frame 0 warm-up)
mean frame = 133,302us ~7.5 fps T1 kernel = 117,260us wall = 117,258us T1@60MHz + 2us libt1/Linux



matmul_8bitraw_short: logical breakdown 10 timed rows + 4 dispatcher-only CSR flips; body rows aggregate 128 repeats
total T1 cycles = 7084.7k total wall = 118,616us libt1 sliver = 538us (@60MHz)



stage	assembly	group	cycles	wall	stage	assembly	group	cycles	wall
	0: vle8.v A	VLE A	12.2k	204us		7: vmv.v.v x128	Shift x128	1764.2k	29,507us
	1: csrwi(vmode=1)	CSR flip	0	0us		8: csrwi V=1 x128	CSR x128	0	8us
	2: vle8.v B^T	VLE B^T	12.1k	202us		9: vrgather.vx x128	Gather x128	1846.3k	30,874us
	3: csrwi(vmode=0)	CSR flip	0	0us		10: csrwi V=0 x128	CSR x128	0	8us
	4: vmv.v.i v7	Init	13.6k	228us		11: vmul.vv x128	Mul x128	633.8k	10,667us
	5: vmv.v.i v28	Init	13.6k	228us		12: vredsum.vs x128	Reduce x128	1192.8k	19,983us
	6: vslideup.vi x128	Shift x128	1585.0k	26,520us		13: vse8.v C	VSE C	11.0k	184us